

SCENARIO — physical and operational description

strain composition (C, N, P) · productivity · operating days · reactor or pond type
harvesting recovery · drying · downstream extraction · plant scale

SHARED MASS AND ENERGY INVENTORY

electricity · heat · carbon source · N · P · substrate · water · solvent · land
plus the annual production the plant delivers

built ONCE; both interpretation layers read the same object

TEA LAYER

prices · CAPEX factors · labour
installation and indirect factors
depreciation · discount rate · tax

LCA LAYER

characterization factors · grid mix
biogenic-carbon convention
system boundary

production cost · CAPEX
NPV · IRR · break-even price

gross GWP · biogenic adjustment · net GWP
CED · water · land · eutrophication

GLOBAL SENSITIVITY (Sobol') AND UNCERTAINTY PROPAGATION (Monte Carlo)
one parameter set, sampled once, carried through the whole chain

prices and characterization factors are NOT inputs to the physical inventory

each parameter draw re-enters at the scenario, so the cost and the impact
it produces always describe one and the same plant